

Surprise Quiz – Python Programming (FULL 1 HOUR)

Time: 60 Minutes

Max Marks: 40

Mode: Hand-written → PDF → Google Classroom

 Instructions (Read Carefully)

- All answers must be hand-written
- Dry run tables are compulsory
- Programs without explanation → 0 marks
- Neatness & logical steps carry marks

Section A – Concept Precision ($8 \times 2 = 16$ marks)

 Expected time: 15 minutes

1. List two situations where tuple is better than list.

2. Explain with example:

Why is `for` loop safer than `while` loop?

3. Predict output and explain:

```
a = [1, 2]
```

```
b = a * 2
```

```
b[0] = 100
```

```
print(a)
```

4. What is the logical difference between:

```
if x:
```

```
if x == True:
```

5. What is the difference between:

```
s[2]
```

```
s[2:3]
```

```
for string s = "HELLO"?
```

6. Why is this bad programming practice?

```
if x == 10:
```

```
    print("A")
```

```
if x > 5:
```

```
    print("B")
```

7. Why does this code never reach else?

if 5 > 2:

print("A")

elif 5 > 1:

print("B")

else:

print("C")

8. What happens if indentation is removed in Python? Why?

Section B – Dry Run & Trace (2 × 8 = 16 marks)

 Expected time: 25 minutes

9. Dry run the following code and write the final output:

lst = [2, 4, 6, 8]

i = 0

total = 0

while i < len(lst):

if lst[i] > 4:

```
    total += lst[i]
    i += 1

print(total)
```

The dry run of this code should be in this format

Table columns:

i | lst[i] | condition | total

10. Dry run and explain:

```
s = "level"
```

```
i = 0
```

```
j = len(s) - 1
```

```
while i <= j:
```

```
    print(s[i], s[j])
```

```
    i += 1
```

```
    j -= 1
```

The dry run of this code should be in this format

Table columns:

i | j | s[i] | s[j] | printed

Section C – Logic Writing (2 × 4 = 8 marks)

 Expected time: 20 minutes

11. Write a program that:

- Takes a list of numbers**
- Counts how many are positive, negative, and zero**
- Uses only one loop**
- Uses if-elif-else**

Explain logic in words.

12. Write a program using nested for loop to generate the pattern:

1

12

123

1234

Explain how the inner loop works.